

The Blind Spot Paradox

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When a self-repairing model destroys the evidence its own monitor needs

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The Blind Spot Paradox. When a self-repairing model destroys the evidence its own monitor needs. Four students of the CentraleSupélec research track, supervised by Raphaël Minato.

The Blind Spot Paradox

└ Two machines, and only one of them talks to you

Picture two machines. The first one predicts something, learns as it goes, and repairs itself when the world stops matching what it learned. The second one is a watchdog. It reads the first one's mistakes and raises an alarm once they pile up. That alarm is the only thing a human ever sees: it is what triggers the audit, the retraining, the pause. Nobody watches the model directly. The watchdog does.

Two machines, and only one of them talks to you

- A **model** predicts, learns as it goes, and **repairs itself** when the world stops matching what it learned.
- A separate **watchdog** reads the model's mistakes and raises an alarm once they pile up.
- The alarm is the only thing a human ever sees. It is what triggers the audit, the retraining, the pause.

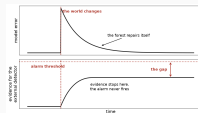
Nobody watches the model. The watchdog does.

The Blind Spot Paradox

└ The repair destroys the evidence

Here is what goes wrong. The world changes, so the model starts making mistakes: that is the top curve. The watchdog needs those mistakes. It adds them up and calls for help once the pile is high enough. But the model repairs itself, the mistakes stop, and the pile stops growing too early. The repair has erased the evidence. The better the model fixes itself, the more surely the alarm stays silent.

The repair destroys the evidence



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└ It is the normal case, not a corner case

It is the normal case, not a corner case

3 alarms
out of 2 000 runs

re-running the source paper's own benchmark, here, on our machines

- One research field measures how fast a model recovers. Another measures how fast an alarm fires.
- Neither has measured them against each other.

This is not a corner case. We re-ran the benchmark of the paper we are studying, and over two thousand runs the alarm fired three times. Now, why had nobody caught this? Because two separate research fields are involved. One studies models that repair themselves and measures how fast they recover. The other studies alarms and measures how fast they fire. Neither had measured the two against each other. That gap is our subject.

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└ What we found: the clock was wrong

To study a race you need a stopwatch, and that is where we found something. The model is a forest of ten trees, and it repairs itself by replacing them one at a time. The date everyone uses to say it has recovered is the moment the very first tree is replaced: one out of ten. Wait instead for three quarters of the forest, and you wait roughly ten times longer. The accepted stopwatch does not time a repair. It times the start of one, always too early.

What we found: the clock was wrong

called "repaired"
by the literature

86 steps

three quarters of
the forest renewed

1225 steps

0 200 400 600 800 1000 1200 1400
time after the change (steps)

median over 100 runs at $\Delta t = 0.24$, $W = 10$

The accepted date marks the first tree out of ten, not a repaired forest.

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└ Where we stand

Where we stand. The paradox is real and reproducible, and we can now say exactly when the evidence falls short of what the alarm needs. The measure the field relies on is biased, and always in the same direction, so any conclusion drawn from it inherits that bias. What remains is to watch the model and the watchdog on one common clock, and turn this into advice a practitioner can actually use. Thank you.

- The paradox is **real and reproducible**, and we can now say exactly when the evidence falls short.
- The measure the field relies on is **biased, always the same way**. Any conclusion drawn from it inherits that bias.
- **Next**: watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.